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**Academic Trimester:** Spring 2026

**Session:** Summer 2022

**Batch/Section:** 28th\_A(day)

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**Department name:** Computer Science & Engineering

**Faculty Name:** Muhammad Anwarul Azim

**University name:** University of Chittagong.

**Course No.:** CSE 423

**Course Title:** Simulation & Modeling.

**Course Credits:** 3.00

**Assignment No.:** 02

**Online Google Form Submission Date:** 05 May,2026

**Total Count of Students in this course/class/batch:** 61

**Total Count of Exercises (or questions or problems):** 197

**Chapter No. :** 05

**Chapter Name:** Statistical Models in Simulation

**The problem (question) serial number:** 93

**The problem (question) Description:** Arrival of vehicles to a gas filling station is Poisson distributed with average 6 per hour. What is the probability of arriving more than 2 vehicles in the next hour

Answer: (93)

Given that, mean,  $\alpha = 6$  per hour

Formula: poisson distribution

$$P(X=x) = \frac{e^{-\alpha} \cdot \alpha^x}{x!}, \quad x=0,1,2,3,\dots$$

we have to find out  $\rightarrow$

$$P(X > 2) = ?$$

$$P(X > 2) = 1 - [P(X \leq 2)]$$

$$P(X \leq 2) = [P(X=0)] + [P(X=1)] + [P(X=2)]$$

$$= \frac{e^{-6} \cdot 6^0}{0!} + \frac{e^{-6} \cdot 6^1}{1!} + \frac{e^{-6} \cdot 6^2}{2!}$$

$$= 0.002478 + 0.01487 + 0.04461$$

$$= 0.061958$$

$$P(X > 2) = 1 - [P(X \leq 2)]$$

$$= 1 - 0.061958$$

$$= 0.9380$$

$$= 0.938 \times 100\%$$

$$= 93.8\%$$

The probability of arriving more than 2 vehicles in the next hour is 93.8%.